EchoChain

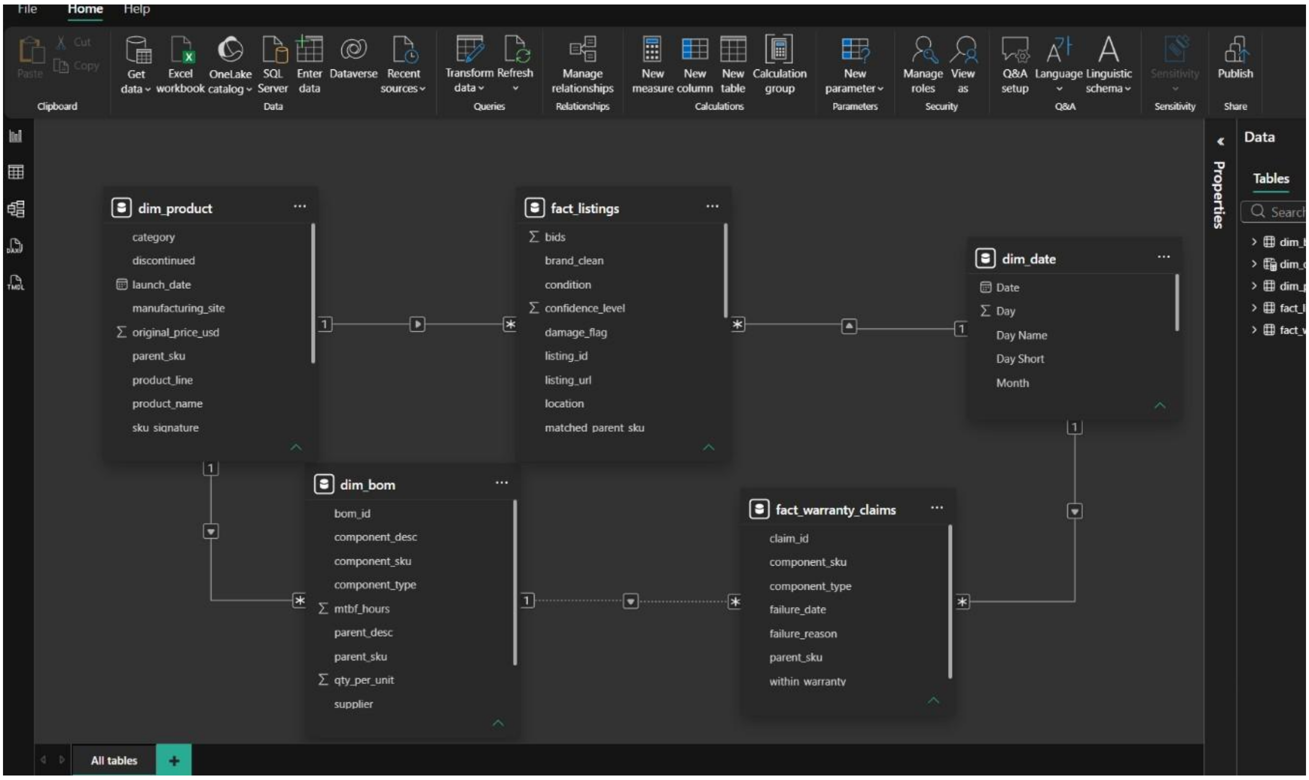
Circular Economy & Secondary Market Lifecycle Analytics

Purpose

Echo Chain connects internal manufacturing information, warranty failures, and secondary-market listings so a manufacturer can understand what happens to a product after sale and identify profitable refurbishment opportunities.

1. Power BI Data Model

The model contains five core tables: product, marketplace listings, BOM/components, warranty claims, and date.



-> dim_product identifies products; dim_bom describes their components; fact_warranty_claims records failures; fact_listings records secondary-market listings; dim_date supports time analysis.

2. Simple Connection Architecture

Think of the model as two paths that meet at the product level.

RELIABILITY / MANUFACTURING PATH

dim_product → dim_bom → fact_warranty_claims

Meaning: Product → What is inside it? → What is breaking?

SECONDARY-MARKET PATH

dim_product → fact_listings dim_date

Meaning: Product → What is it worth used? → When was it listed?

BUSINESS ANALYTICS

Resale value + depreciation + component reliability + repair/refurbishment potential →

Circularity Score

3. dim_product — Product Master

This is the main reference table for products. It answers: **"What product are we talking about?"** One product can connect to many marketplace listings and many BOM rows.

4. fact_listings — Secondary-Market Listings

This is the marketplace fact table collected by the Scrapy layer. It answers: **"What is this product worth after sale?"**

5. dim_bom — Bill of Materials

BOM means **Bill of Materials**. It tells EchoChain what components are inside each product. One product can have many component rows.

6. fact_warranty_claims — Warranty Failures

This table records component-level warranty claims. It answers: **"What is failing, why is it failing, and when?"**

7. dim_date — Calendar Dimension

dim_date is the shared calendar used for time-based analysis such as monthly resale trends and warranty failures over time.

8. Relationships in the Screenshot

dim_product (1) → fact_listings (*)
One product can have many marketplace listings.

dim_product (1) → dim_bom (*)
One product can have many BOM/component rows.

dim_date (1) → fact_listings (*)
One date can have many marketplace listings.

dim_date (1) → fact_warranty_claims
(*) One date can have many warranty
claims.

dim_bom (1) → fact_warranty_claims (*)
The screenshot shows this relationship as dotted/inactive. Check this relationship before building DAX measures; an inactive relationship may require USERELATIONSHIP or a revised model.

9. End-to-End EchoChain Example

Suppose the manufacturer has laptop SKU **LAP001**.

Step 1 — Product

dim_product says LAP001 is a laptop with an original price of $1,200.

Step 2 — Components

dim_bom shows that LAP001 contains a motherboard, display, RAM, SSD, battery and other components.

Step 3 — Failures

fact_warranty_claims shows that the motherboard fails frequently while the display has fewer failures.

Step 4 — Used-market value

fact_listings shows used LAP001 listings around $500–$550. Strong condition and valuable components can indicate good recovery potential.

Step 5 — Decision

The executive compares failure risk with retained resale/component value. If the display is valuable and the product is economically repairable, a buy-back and refurbishment program may be attractive.